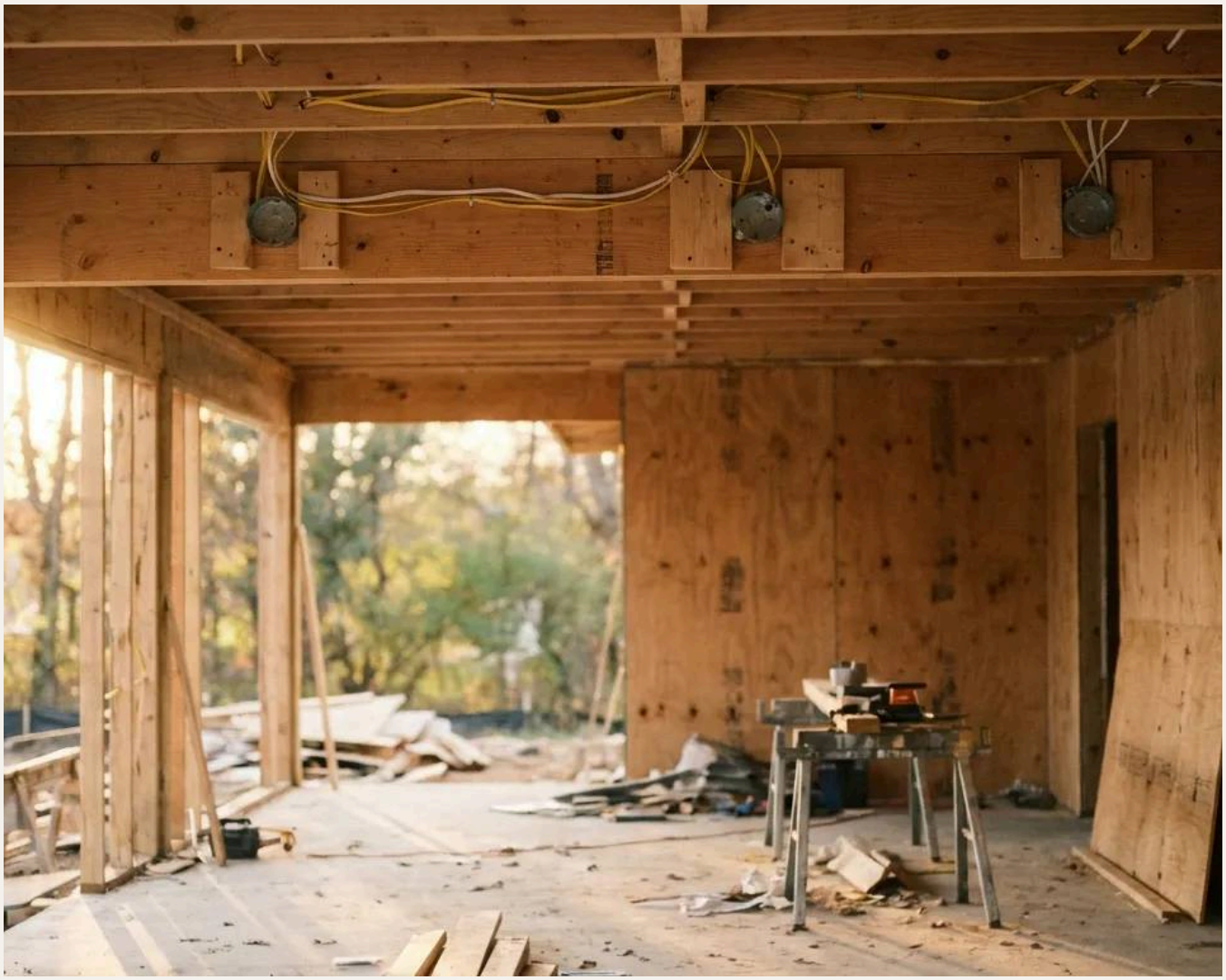
BROMIC
HEATING

BUILDER / INSTALLER

Pre-wire for outdoor heat: the builder's rough-in guide

Rough-in guide for patio heaters: circuits and amps, 125% sizing, junction boxes, blocking, recess framing, control runs and gas stub-outs.



EDITION
October 2026

BROMIC HEATING
bromic.com

Pre-wire for outdoor heat

The builder's rough-in guide. Designed from the start, not added later.

Rough in a dedicated branch circuit for each electric heater (or heater group, per the controller), sized from I = W ÷ V with conductors at 125% of the load; a junction box and solid blocking at every heater, pole or recess location; a home run from each heater to the controller location; a box at the wall switch; and, for built-in gas, a stubbed and capped line with an accessible shut-off. Decide all of it before framing, while the ceiling is open.



Covered outdoor kitchen and bar with flush ceiling heaters

Why decide before framing?



Fewer than one in four homeowners prioritize heating at planning[1]. The rest ask for it after drywall, soffits and stone are done. That means opening a finished ceiling, surface conduit, a heater placed where the wire reaches rather than where the table sits, and a callback.

DECIDED AT	WHAT IT COSTS
Design / pre-framing	A line on the electrical plan; blocking and boxes at rough-in
After framing, before close-in	A change order; extra trip for the electrician
After finish	Opened ceilings or walls, patching, repainting, surface conduit, often no recess option

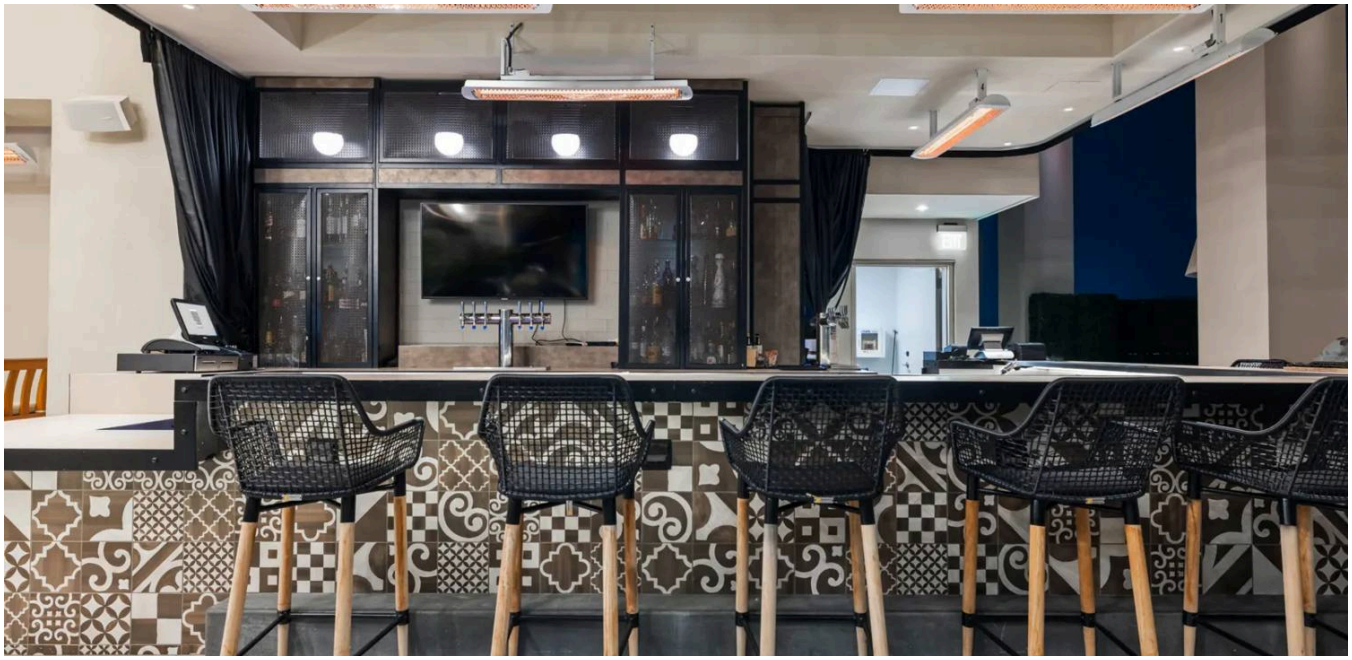
Sell it at the options stage. A pre-wire package (circuits, boxes, blocking, switch box) can be offered even when the heaters are bought later. See the builder programme and Good–Better–Best guide (A15).

The decision timeline



STAGE	DECIDE	WHO
Plans	Heated zones, heater type and count, mount type, fuel	Builder, designer, Bromic design service
Pre-framing	Recess openings, blocking, ceiling build-up	Framer, builder
Electrical rough-in	Circuits, boxes, control runs, switch location, panel capacity	Licensed electrician
Gas rough-in	Line size, stub-out, shut-off	Licensed gas fitter
Rough inspection	Inspect before close-in	AHJ
Trim-out	Heaters, recess kits, controllers, commissioning	Electrician, gas fitter

What to rough in, by product



PRODUCT	FUEL / RATING	ROUGH-IN	MOUNT PROVISIONS
Platinum Smart-Heat Electric	2300 / 3400 / 4500 W	Dedicated circuit(s) to controller; heater junction box	Recessed (recess kit), ceiling or wall; blocking
Eclipse Smart-Heat Electric		Circuit to controller; ceiling box at pole	Single pole 8", 24" or 48"; double or curved pole; blocking for pole
Eclipse Smart-Heat Freestanding Electric			None; level surface
Tungsten Smart-Heat Electric — Coastal / Select (coming soon)		As Platinum, pending install manual	Low-clearance recessing, ceiling, wall, single-post
Tungsten Gas (built-in), Platinum Gas		Gas stub-out and shut-off;	Ceiling, wall; ceiling poles, heat deflectors
Tungsten Smart-Heat Freestanding Gas	38,000 Btu/h; natural gas or propane	Propane: none. Natural gas:	None

Product facts: Bromic Lookbook 2026[2] and Planning Guide 2021[3]. Always build to the current installation manual for the model.

How many amps? The maths



Step 1. Current per heater. I = W ÷ V.

Step 2. Continuous load. Fixed electric space-heating equipment is treated as a continuous load. NEC Article 424 requires branch-circuit conductors for it to be sized at not less than 125% of the load, consistent with the continuous-load rules in 210.19(A)(1) and 210.20(A)[54]. Feeder and service calculations follow Article 220.

Step 3. Circuit and breaker. The electrician selects conductor and overcurrent device from the result, the installation manual and the adopted NEC.

HEATER	AMPS AT 208 V	× 1.25	AMPS AT 240 V	× 1.25
2300 W	11.1	13.8	9.6	12.0
3400 W	16.3	20.4	14.2	17.7
4500 W	21.6	27.0	18.8	23.4

Rated wattage divided by voltage. A heater rated at 240 V draws less on 208 V; use the rating of the model specified. Commercial sites may also be offered 277 V[3].

Worked example: three 3400 W heaters on 240 V.

- Per heater: 3400 ÷ 240 = 14.2 A; × 1.25 = 17.7 A.
- All three on one controller output: 42.5 A; × 1.25 = 53.1 A minimum conductor ampacity — too much for one typical circuit, so split them.
- Panel: add 10.2 kW to the load calculation.

DIAGRAM: Single-line diagram: panel → dedicated breakers → Affinity controller (one channel per zone) → junction boxes at heaters; wall switch and Wi-Fi shown; conductor sizes left blank for the electrician.

Where do the boxes and blocking go?

ITEM	POSITION	NOTE
Heater junction box	At each heater location, per manual	
Wall-mount height	Heater ≥ 72" above finished floor	Bromic guidance[3]
Ceiling-mount height	Heater ≥ 96" above finished floor	Clearance to ceiling varies with output[3]
Blocking	Solid blocking between joists at every heater, pole and bracket	Sized for heater and pole weight
Suspension poles	Blocking and box at pole top	For high ceilings[3]
Clearances	To combustibles, ceiling and sides, per manual	Do not interpolate between models
Aim	Over seating, not the floor centre	Coordinate with furniture plan

White heater between pergola louvres

How do I frame for a recess kit?

Recess kits flush-mount Platinum and Tungsten electric heaters into the ceiling cavity[3]. They need to be in the framing plan.

DIMENSION	VALUE
Rough opening (L × W)	
Minimum cavity depth	
Clearance above kit to combustibles	
Low-clearance recessing (Tungsten Coastal / Select, coming soon)	
Ceiling finish thickness allowed	

- Run joists or add headers to frame the opening square and level.
- Keep insulation, vapour barriers and other services clear of the kit per the manual.
- Mark openings on the framing plan and on the ceiling before close-in.

DIAGRAM: Reflected ceiling plan and section of a recess-kit opening between joists: header, blocking, junction box, cavity depth and clearance dimensions shown as callouts awaiting verified values.

Tungsten Coastal recessed flush into a white ceiling

Controls: what to run to the switch and controller



Controls are part of the system. Rough them in with the heater circuits.

RUN	FROM → TO	NOTE
Line voltage	Panel → Affinity controller → heaters	One controller channel per zone
Wall switch	Controller → switch box at the threshold	
Controller location	Accessible, protected, within Wi-Fi range	
Integration	Controller → home-automation panel	Control4, Crestron or Lutron; compatibility varies and may need extra hardware[2]
Spare conduit	Heater zone → controller location	Allows later zones or integration without opening finishes

Specify the matching control for every heater: an Affinity Smart-Heat On/Off or Dimmer controller (Platinum, Tungsten electric) or Eclipse controller (Eclipse), plus a wall switch. Wi-Fi and Bromic Smart-Heat app features require compatible Affinity Smart-Heat Controls, sold separately[2]. Check Wi-Fi signal at the controller location before close-in.

Gas stub-outs for gas models



- **Licensed gas fitter only.** Size, route and test the line to ANSI Z223.1/NFPA 54[6] and the local code.
- **Size for the connected load.** Total Btu/h of all appliances on the line, run length and supply pressure.
- **Stub out and cap** at each heater location; provide an accessible shut-off.
- **Mount provisions:** heat deflectors reduce the clearance needed above gas heaters; ceiling poles mount two heaters back to back[3]. Allow for both in the framing.
- **Fuel at order:** natural gas or propane, confirmed before the heater is ordered.

What will the inspector check?

- [ ] Permits pulled for electrical and gas work
- [ ] Circuits match the approved plan and load calculation
- [ ] Boxes accessible, supported and at the planned positions
- [ ] Blocking in place for every heater and pole
- [ ] Recess openings framed to the manual
- [ ] Gas line pressure-tested and capped; shut-offs accessible
- [ ] Installation manuals on site for the inspector
- [ ] Final: heaters listed and marked (Intertek, UL or CSA)[3]; clearances met; controls working

This guide is general guidance, not an engineering or code determination. Confirm requirements with the local building authority. Electrical work by a licensed electrician to the adopted NFPA 70[4]; gas work by a licensed gas fitter to ANSI Z223.1/NFPA 54[6]; both to the manufacturer's installation instructions.

Black heater on a timber cabana wall at dusk

Printable rough-in checklist



Before framing

- [ ] Heated zones and seating drawn
- [ ] Heater model, count, fuel and mount type chosen per zone
- [ ] Recess kits and poles identified
- [ ] Framing plan shows openings and blocking

Electrical rough-in

- [ ] Load per heater calculated (I = W ÷ V) at the supply voltage
- [ ] Conductors ≥ 125% of load; breakers per manual and NEC
- [ ] Panel capacity checked; load added to Art. 220 calculation
- [ ] Junction box at every heater; heights ≥ 72" wall, ≥ 96" ceiling
- [ ] Home runs to the Affinity controller location
- [ ] Wall-switch box at the threshold
- [ ] Spare conduit to controller location
- [ ] Wi-Fi coverage confirmed at controller

Gas rough-in (gas models)

- [ ] Line sized for total Btu/h by licensed fitter
- [ ] Stub-outs capped at each location; shut-offs accessible
- [ ] Pressure test passed

Before close-in

- [ ] Rough inspection passed
- [ ] Photos of every box and opening, with dimensions, filed with the job

Electrician hand-off sheet

JOB	
Address / lot	
Builder contact	
Supply voltage	208 V / 240 V / 277 V
Panel location and spare capacity	

TAG	ZONE	MODEL	WATTS	VOLTS	AMPS (W ÷ V)	×1.25	CIRCUIT / BREAKER	MOUNT	HEIGHT AFF
H1									
H2									

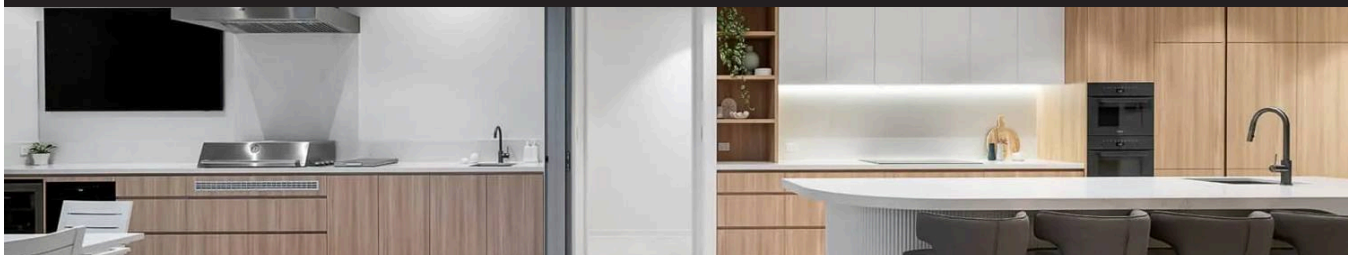
TAG	ZONE	MODEL	WATTS	VOLTS	AMPS (W ÷ V)	×1.25	CIRCUIT / BREAKER	MOUNT	HEIGHT AFF
H3									
H4									
CONTROLS									
Affinity controller(s) and channels									
Controller location									
Wall-switch location									
Integration platform (Control4 / Crestron / Lutron / none)									
Wi-Fi confirmed at controller									Y / N
NOTES									
Recess openings (size, location)									
Blocking									
Gas stub-outs (if any)									
Installation manuals attached								Y / N	

Large pergola patio with outdoor kitchen and dining

Book a design consult and get the layout, heater schedule and controls bundle before framing.

NEXT STEP

Book a design consult



Sources

1. **Bromic Heating / Atomik Research**, Outdoor Living Trends 2026 (US Homeowner Survey) (2026)
2. **Bromic Heating**, Lookbook 2026 (2026)
3. **Bromic Heating**, The Official Outdoor Heating Planning Guide, US v1.2 (2021)
4. **National Fire Protection Association**, NFPA 70, National Electrical Code (2023). https://www.nfpa.org/codes-and-standards/nfpa-70-standard-development/70
5. **EC&M**, Article 424: Fixed Electric Space Heating (2021). https://www.ecmweb.com/national-electrical-code/code-basics/article/20888653/article-424-fixed-electric-space-heating
6. **National Fire Protection Association**, NFPA 54, National Fuel Gas Code (ANSI Z223.1) (2024). https://www.nfpa.org/codes-and-standards/nfpa-54-standard-development/54

Confirm every installation with your local building authority and a licensed electrician or gas fitter. Wi-Fi and app features need compatible Affinity Smart-Heat controls, sold separately. © 2026 Bromic Heating · bromic.com